



Atharva Bhosale

BTech
Mechanical Engineering
Indian Institute Of Technology, Tirupati

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Github LeetCode
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (ME)	Indian Institute of Technology, Tirupati	7.76	2024-2028
Senior Secondary	Vidya Valley North Point Jr College, Chinchwad HSC (MH) Board	86.33%	2024
Secondary	Podar International School, Chinchwad CBSE Board	95.8%	2022

EXPERIENCE

- Techmaniacs, IIT Tirupati** Sept. 2024 - Aug. 2026
Robotics Software Team Member On-Campus
 - Developed Python and ROS2-based software for SLAM, teleoperation and hardware-interface systems on a TortoiseBot.
 - Processed and analyzed sensor and robot-state data for localization, navigation and system debugging across simulated and physical environments.
 - Built and calibrated a drone system and developed an ArduPilot-based geofencing program for region-constrained flight.
 - Conducted robotics workshops, communicating technical concepts and guiding students through Python, ROS2 and simulation workflows.

PROJECTS

- Autonomous Drone Precision Landing** Sept. 2026
Data-driven analysis of controller performance across 50 autonomous landing trials. GitHub
 - Tools:** Python, OpenCV, PyBullet, PPO, PID, Kalman Filter, Pandas
 - Built an ArUco + Kalman filter vision pipeline to track dynamic target positions in PyBullet simulation.
 - Evaluated PID and PPO-based control across 50 trials, improving landing success from 34% to 68% and reducing touchdown error by 28.7%.
- Ford Car Price Exploratory Data Analysis** Aug. 2026
Exploratory data analysis of 17,900+ Ford car listings to uncover pricing trends. GitHub
 - Tools:** Python, Pandas, NumPy, Seaborn, Matplotlib
 - Analyzed 17,966 records across 9 features using Pandas and Seaborn to discover key price drivers.
 - Executed data quality checks, identifying corrupt outliers in manufacture year and MPG features.
- Stock Price Analytics & Forecasting** May 2026
Time-series analysis and forecasting of NSE stock prices using engineered market features. GitHub
 - Tools:** Python, Scikit-Learn, Pandas, NumPy, Matplotlib
 - Engineered 5+ time-series features including lags, moving averages and rolling volatility across multiple NSE stocks.
 - Trained Linear Regression models and evaluated 30-day forecasts using MAE and RMSE against a naive baseline.

TECHNICAL SKILLS

- Programming:** Python, SQL, C/C++, JavaScript
- Data Analytics:** Pandas, NumPy, Data Cleaning, EDA, Statistical Analysis
- Machine Learning:** Scikit-learn, Machine Learning, Reinforcement Learning, OpenCV
- Visualization:** Matplotlib, Seaborn, Excel
- Tools:** Git, Docker, Linux
- Robotics & Simulation:** ROS2, Gazebo, PyBullet, ArduPilot

KEY COURSES TAKEN

- Data Structures & Algorithms, Introduction to Data Science & Machine Learning, Numerical Analysis

POSITIONS OF RESPONSIBILITY

- Club Head,**IdeaSquare tinkering and innovation hub, IIT Tirupati Present
- Batch Representative,**Career Development & Guidance Council, IIT Tirupati July. 2025 - May. 2026
- Core Member,**TechManiacs Tech Club, IIT Tirupati Aug. 2024 - Aug. 2026
- Core Member,**SLHD Tirutsava-25 Cultural Fest, IIT Tirupati Sept. 2024 - Feb. 2025

ACHIEVEMENTS

- Secured 2nd Position** TechHeist A Capture The Flag Event, Tirutsava-25 2025

CERTIFICATIONS

- Udemy Cert on Nav2 With SLAM And Navigation, TFU|RDF|Rviz|Gazebo, Robot Operating System, Python